INTEGRATED DEVELOPMENTAL SCHOOL

MIOW LMS Performance audit

Quantitative audit: 21 Lighthouse runs + 15 scenarios

Measured 7 October 2026 | Asia/Singapore

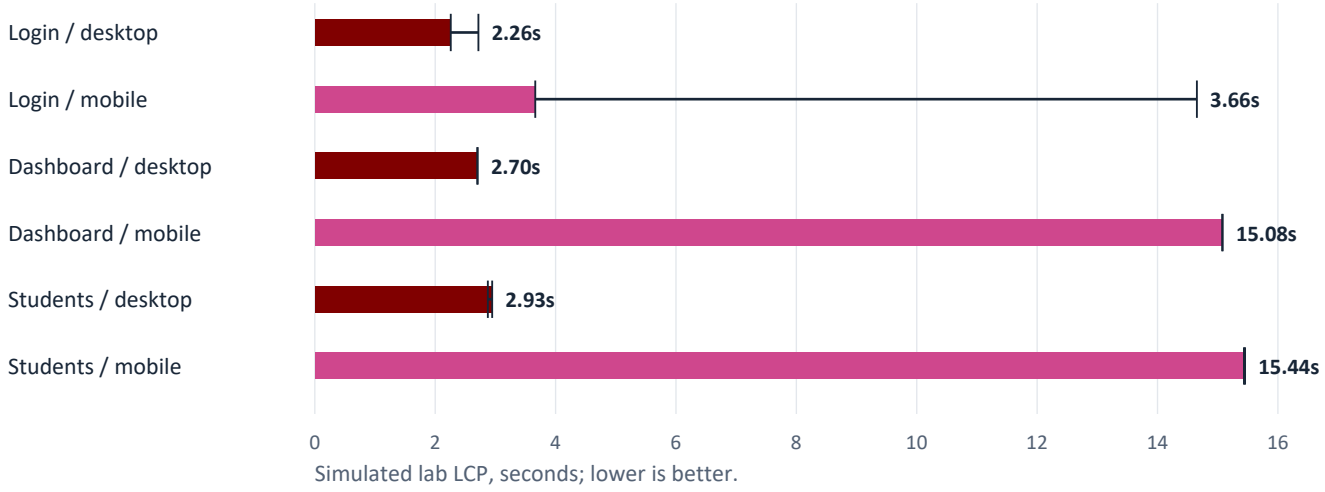
https://miow-lms.ter-ids.online/

Key findings

Mobile performance remains slow. Baseline median LCP is 15.08 seconds for Dashboard and 15.44 for Students; the newer Students deployment measures about 15.10 seconds. Results for the two deployments are kept separate.

A 1.79 MB preloaded script accounts for about 68% of baseline login transfer and remains in the newer build. Splitting it from the entry path is the largest measured payload opportunity.

Baseline LCP: median with the full three-run range



This is a separate follow-up to the first UI/UX PDF. The first report and its inline screenshots remain unchanged. Results here are lab measurements and sampled interactions, not a real-user Core Web Vitals or INP assessment.

Completed 7 October 2026 (Asia/Singapore). This is a separate quantitative performance report, supplementing the earlier UI/UX audit. The first audit and its inline screenshots are unchanged.

Test method and conditions

Lighthouse 13.5.0 with Chrome/154.0.8037.98, run locally in isolated headless Chrome. Host: AMD Ryzen 5 5500 ; 12 logical CPUs; 16 GiB RAM. Eighteen baseline runs: three runs for each of three pages in each device profile. Three additional mobile Students runs tested a later deployment. Pages were tested sequentially. The authenticated tests used the supplied admin account through the normal login form.

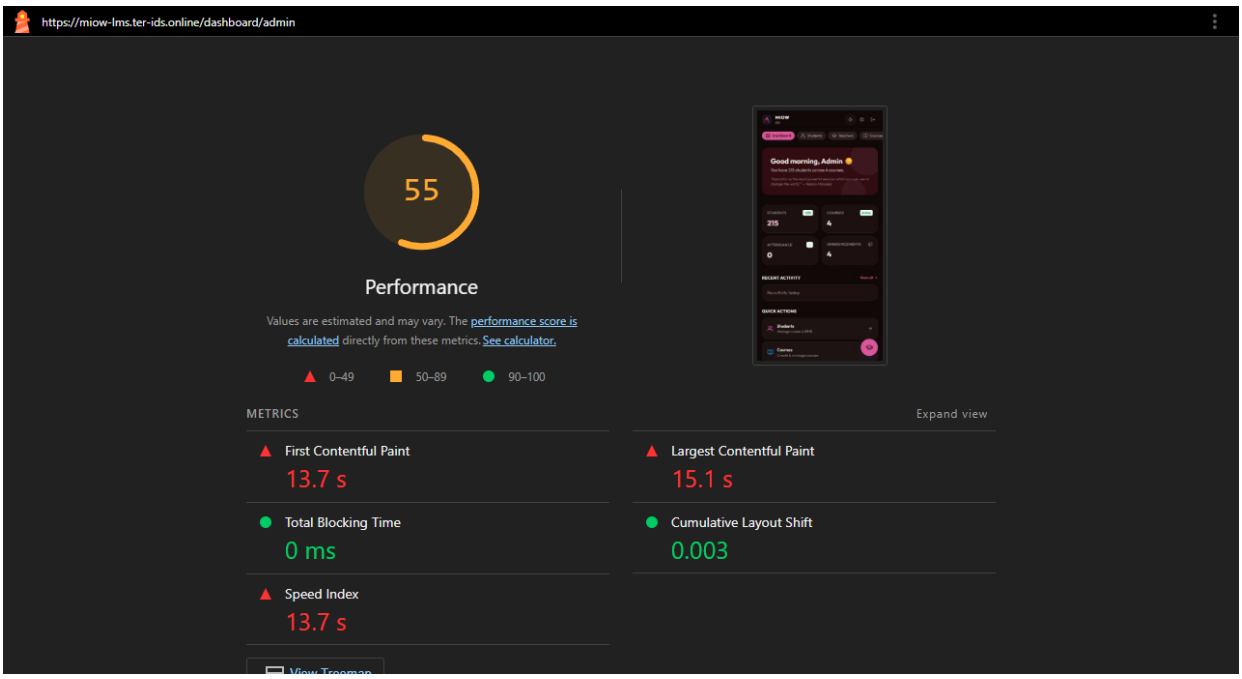
The baseline Lighthouse series ran from 03:03:25 to 03:08:59 Singapore time and all 18 reports loaded **index-DwGbiE0M.js**. During the later interaction checks, the site changed to **index-CQrrc219.js**; the Students route module changed from **dashboard.admin.students-b46lrLH2.js** to **dashboard.admin.students-DJ4qNPqt.js**. These asset fingerprints distinguish the observed deployments. Baseline and follow-up figures are reported separately. The newer login and dashboard deployment was not re-audited with Lighthouse.

Before each final Lighthouse run, the page navigated to blank and the HTTP cache was cleared. Every final report was checked for zero disk/memory-cache responses. Authentication storage was retained without exporting it. Tests used isolated temporary Chrome profiles across batches. DNS, TLS, JIT, server, and CDN caches may remain warm. These are cold HTTP-cache lab tests, not first-ever network connections.

Profile	Viewport	Network simulation	CPU
Desktop	1350 x 940, DPR 1	40 ms RTT; 10,240 Kbps	1x
Mobile	412 x 823, DPR 1.75	150 ms RTT; 1,638.4 Kbps	4x slowdown

Lighthouse FCP, LCP, Speed Index, and TBT below use simulated throttling. Document TTFB is observed on the unthrottled test connection. These columns describe different measurement conditions and should not be compared as one trace. Each reported median is calculated independently. Earlier preliminary runs with inconsistent cache use are excluded from the final results and retained separately for transparency.

Representative baseline mobile dashboard Lighthouse report



Inline evidence: dashboard-mobile-2. An individual run; the results tables report independent three-run medians and ranges.

Median Lighthouse results

FCP is first contentful paint; LCP is largest contentful paint; TBT is total blocking time; CLS is cumulative layout shift. Speed Index tracks how quickly visible content appears. Lower values are better for these metrics; a higher performance score is better.

Page / device	Score	FCP (s)	LCP (s)	Speed Index (s)	TBT (ms)	CLS
Login / desktop	76/100	2.26	2.26	2.26	0	0.001
Login / mobile	81/100	3.66	3.66	3.66	0	0.000
Dashboard / desktop	71/100	2.46	2.70	2.46	0	0.002
Dashboard / mobile	55/100	13.65	15.08	13.65	0	0.003
Students / desktop	68/100	2.43	2.93	2.71	29	0.073
Students / mobile	52/100	13.12	15.44	13.12	166	0.080

Variability and transfer cost

Page / device	Score range	LCP range (s)	TTFB (ms)	Transfer (MiB)	Requests
Login / desktop	70-76	2.26-2.72	56	2.53	81
Login / mobile	55-81	3.66-14.65	57	2.53	81
Dashboard / desktop	71-71	2.70-2.71	53	2.55	87
Dashboard / mobile	55-55	15.07-15.08	57	2.55	87
Students / desktop	67-68	2.87-2.95	53	2.63	208
Students / mobile	51-53	15.44-15.45	55	2.63	208

Transfer includes response overhead as reported by Lighthouse, not just decoded file sizes. Request counts include background and application requests observed within the audit window. Different profile totals can reflect when dynamic requests finish. Scores and LCP ranges preserve all three valid runs; no slow result was discarded.

The mobile login LCP ranged from 3.66-14.65 seconds despite zero cached responses. The three-run median should be read alongside that variability. Browser compilation, connection reuse, and server/CDN warming were not independently isolated; a fresh-profile repeat series would help explain the difference.

Applied-throttle usability checks

Fifteen scenarios applied Chrome DevTools Protocol throttling. Ten belong to the baseline deployment and five to the newer mobile Students deployment. The table below contains only baseline samples: three per combination except mobile Students, which has one observation before the deployment changed. Dashboard ready means the real 215-student / 4-course summary and quick actions were present; Students ready means the search input and the real 215-of-215 roster count were present. These definitions measure useful content, not complete network idleness.

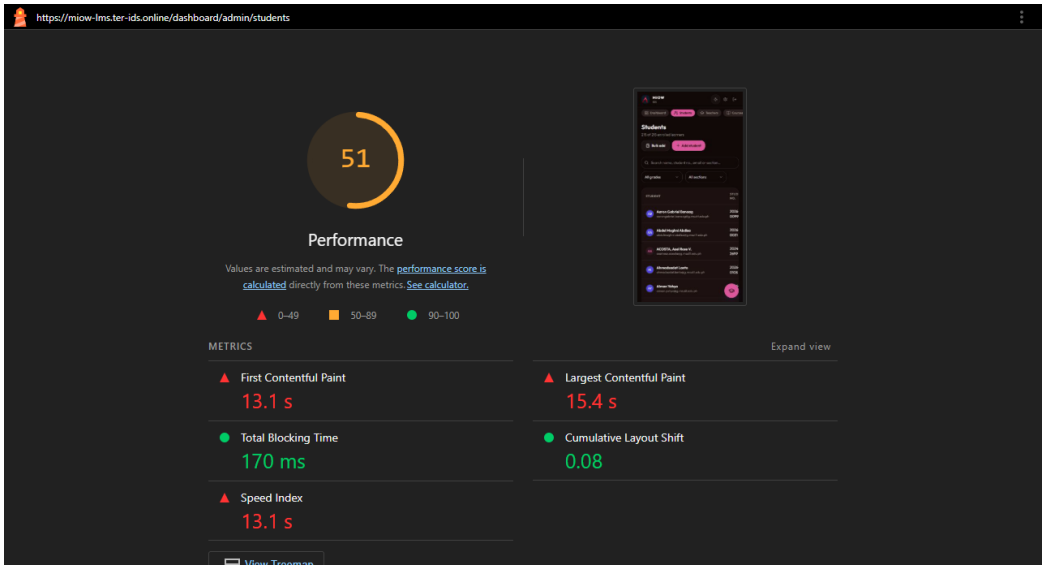
Desktop applied settings: 40 ms request latency, 10,240 Kbps download/upload, CPU 1x. Mobile applied settings use Lighthouse slow-4G request-level adjustments: 562.5 ms request latency, 1,474.56 Kbps download, 675 Kbps upload, CPU 4x; 412 x 823 viewport at DPR 1. These applied scenarios are not numerically identical to Lighthouse simulation.

Page / device	N	Ready (s)	Ready range (s)	Search (ms)	Restore (ms)
Dashboard / desktop	3	5.22	5.05-5.62	-	-
Dashboard / mobile	3	19.56	19.27-19.57	-	-
Students / desktop	3	4.57	4.37-5.65	141	103
Students / mobile	1	20.31	20.31-20.31	528	490

Search measures typing the 19-character no-match query and seeing its result after two animation frames. Restore measures Select All + Backspace and seeing the complete roster after two frames. They include scripted input delivery and are scenario timings, not pure browser event latency.

Browser Event Timing, desktop: median of per-run maximum sampled interaction durations = 80 ms (range 72-80 ms). Durations are quantized by the browser. These short scripted samples do not establish real-user INP. Browser Event Timing, mobile: median of per-run maximum sampled interaction durations = 368 ms (range 368-368 ms). Durations are quantized by the browser. These short scripted samples do not establish real-user INP.

Representative baseline mobile Students Lighthouse report



Inline evidence: baseline students-mobile-2. Simulated Lighthouse metrics are separate from applied-throttle search timings.

The single baseline mobile Students interaction sample is not a repeated estimate. The newer deployment has five observations below.

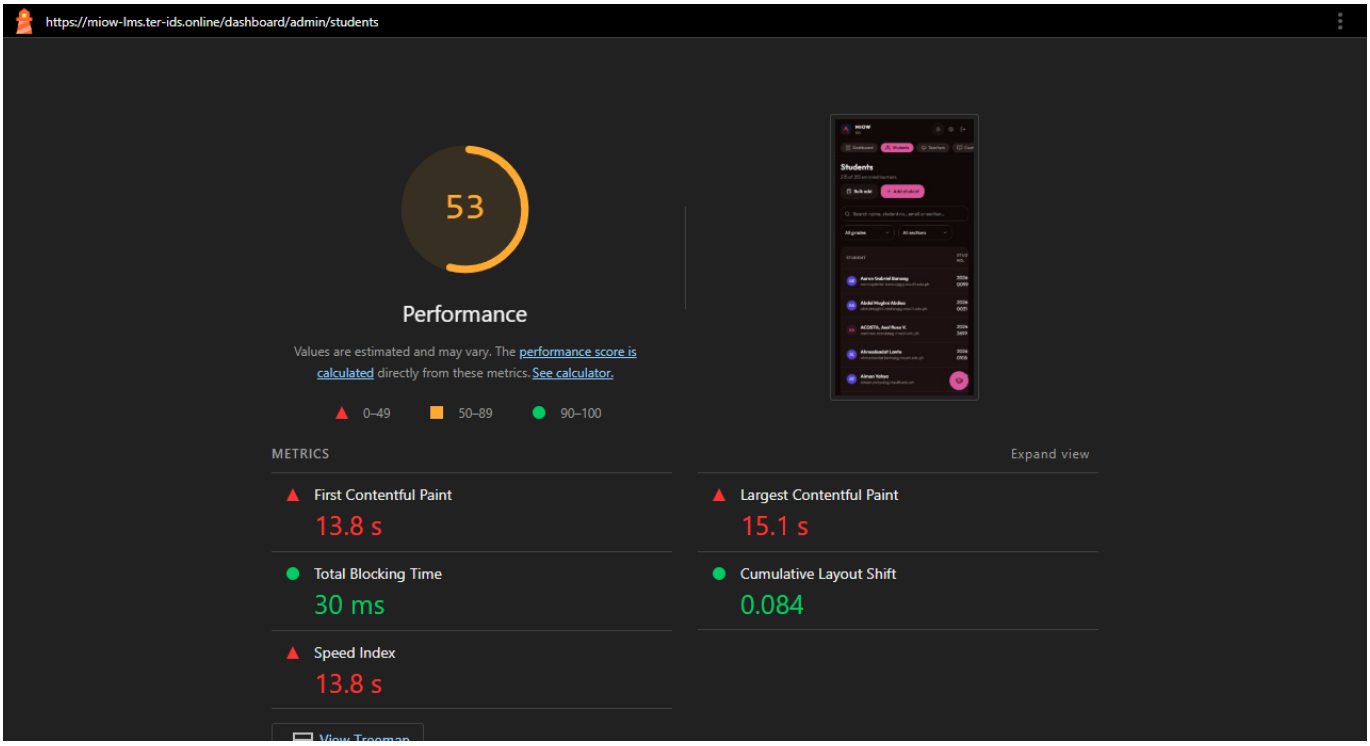
Follow-up on the newer Students deployment

Three cold-cache mobile Lighthouse runs on **index-CQrrc219.js** produced a median score of 53/100 (range 53-53), LCP 15.10 s (range 15.08-15.12), TBT 18 ms, and CLS 0.084. Median transfer was 2.56 MiB across 107 requests. These results describe a different deployment from the baseline table.

Five applied mobile Students scenarios on the newer deployment produced useful-content readiness of 19.25 s (range 18.87-19.37), search completion 649 ms, and restore completion 239 ms. The median per-run maximum sampled Event Timing was 152 ms. These remain scripted lab samples, not field INP.

The newer roster rendered about 1149 DOM elements and 23 images at readiness, with 23 images using native lazy loading. The earlier mobile observation had 4,638 elements and 124 images with none using native lazy loading. This confirms a substantial rendering change, while the large Shiki preload is still present. The measurements do not isolate which code changes caused each timing difference.

Newer mobile Students Lighthouse report



Inline evidence: post-deployment/students-mobile-2. The newer report belongs to a separate deployment and is not pooled into the baseline medians.

Measured bottlenecks and fixes

1. Split the large preloaded script from the entry path. The login request list contains **shiki-DFkX5emM.js**: 1,793,432 transfer bytes (1.71 MiB) and 10,252,656 decoded bytes (9.78 MiB). It is a high-priority preload and accounts for about 68% of the login transfer. The Shiki filename suggests syntax-highlighting functionality; its import purpose should be confirmed in source. Load the feature only when a worksheet/editor requires it, and import only required languages/themes. Removal savings are a payload opportunity, not a measured before/after LCP improvement.



Large preloaded script: 68% of login transfer Other resources: 32%

Login representative desktop run: 73 script requests, 2.47 MiB JavaScript transferred; 0 image requests, 0.0 KiB image transfer. Dashboard representative desktop run: 74 script requests, 2.47 MiB JavaScript transferred; 1 image requests, 0.7 KiB image transfer. Students representative desktop run: 75 script requests, 2.47 MiB JavaScript transferred; 123 image requests, 81.8 KiB image transfer.

2. Profile the data path as well as the document response. Fast HTML TTFB does not mean the admin data is ready. Representative dashboard and roster runs include opaque **_serverFn/** fetches; inspect these handlers, query timing, region placement, duplicate calls, and waterfall dependencies before attributing the delay to one backend component.

Dashboard representative desktop run: 5 completed Fetch requests; duration range 1.24-2.29 s on the unthrottled audit connection. These are request durations, not pure server execution times. Students representative desktop run: 3 completed Fetch requests; duration range 1.47-2.78 s on the unthrottled audit connection. These are request durations, not pure server execution times.

3. Verify the newer roster rendering improvements. The baseline representative Students report made 208 requests, including 123 image requests. The baseline mobile observation had 4638 DOM elements and 124 images, with 0 using native lazy loading. The follow-up already shows fewer elements and lazy-loaded images. Keep the smaller rendered roster, verify pagination/filter behavior, and reserve row dimensions during loading. Images are numerous but their bytes are much smaller than the large script bundle; prioritize that bundle for transfer reduction.

4. Reduce render-blocking work and loading movement. Lighthouse identifies blocking stylesheet/font requests and loading-related layout shifts. Review the external Outfit stylesheet, font strategy, and critical CSS. Keep skeleton dimensions consistent with loaded cards/rows. Savings estimates in raw Lighthouse reports are diagnostic projections and are not additive guarantees.

Interpretation and practical limits

All 21 accepted Lighthouse runs completed without runtime failures and with zero cached responses. The network records contain 0 HTTP 4xx/5xx responses across these runs. Browser page-error capture returned 0 entries. This is a sample of the tested flows, not proof that every application path is error-free.

For reference, the good real-user Core Web Vitals thresholds are LCP <= 2.5 s, INP <= 200 ms, and CLS <= 0.1 at the 75th percentile of visits. This audit provides lab measurements and sampled interactions. It does not establish a field Core Web Vitals pass, and TBT is not INP. Add real-user monitoring to assess INP and the distribution of real visitors.

Re-test the same three routes with the same cache and throttling settings after code changes. Keep the median and range for three runs. First target the large entry-path bundle, then data readiness and roster rendering.

Measurement references: Lighthouse performance scoring, Lighthouse throttling methodology, Authenticated Lighthouse setup, Core Web Vitals definitions.